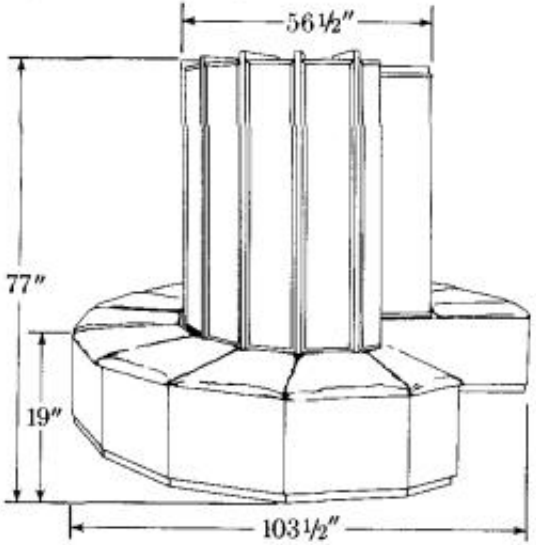
The CRAY-1 Computer System

“The world’s most expensive love-seat”



Fig. 1. Physical organization of mainframe.



- Dimensions
  - Base–103½ inches diameter by 19 inches high
  - Columns–56½ inches diameter by 77 inches high including height of base
- 24 chassis
- 1662 modules; 113 module types
- Each module contains up to 288 IC packages per module
- Power consumption approximately 115 kw input for maximum memory size
- Freon cooled with Freon/water heat exchange
- Three memory options
- Weight 10,500 lbs (maximum memory size)
- Three basic chip types
  - 5/4 NAND gates
  - Memory chips
  - Register chips

A “reasonably trim individual” can gain access to the interior of the machine.

- 12.5 ns clock
- 8 MB internal semiconductor memory
- 4 KB of register storage
- Uses ECL (Emitter-Coupled Logic) throughout
- 115 kW input power
- Simple gates

Memory

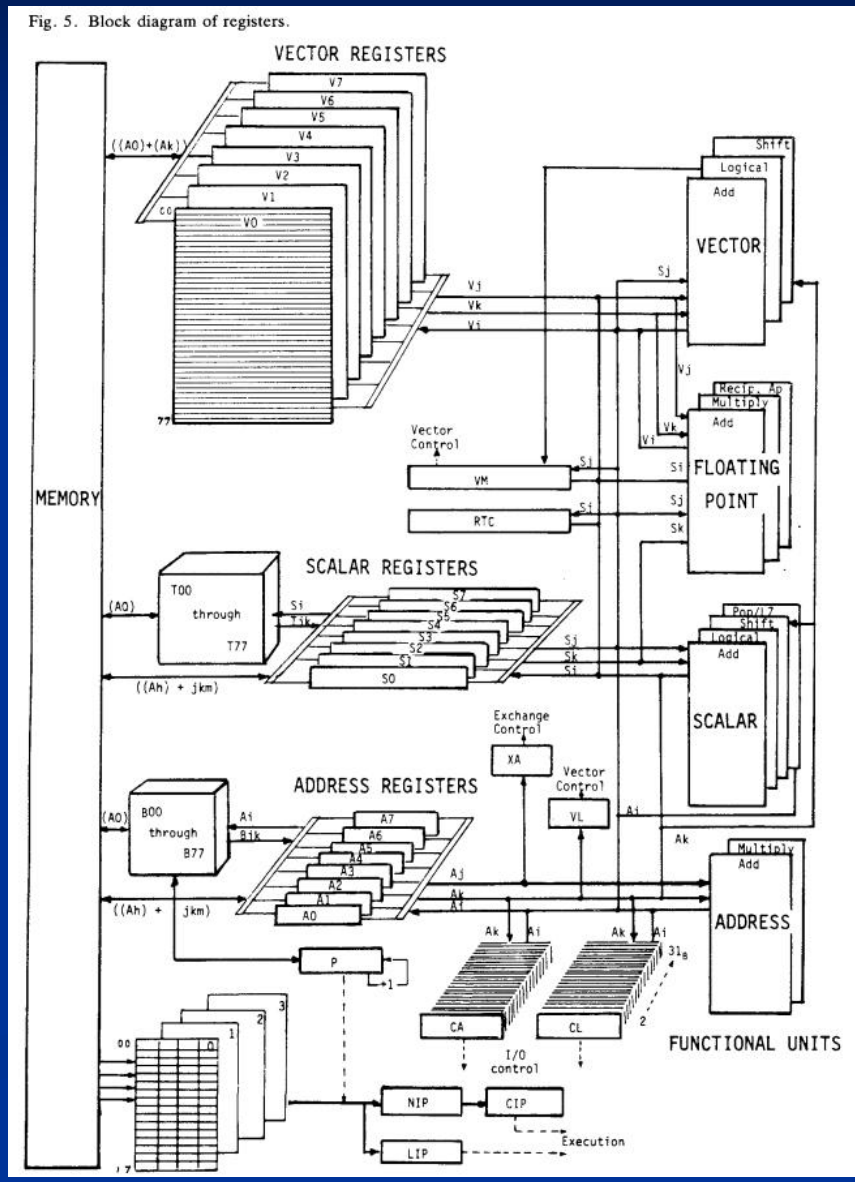
- 16 bank = 16 way interleaved access
- No bank conflicts except on stride lengths of 8 or 16
- 4 clock cycles per access
- Can pull down 16 instructions per cycle
- 1 data word if being placed in registers

Cooling

- Big power + many modules = heat
- Aluminum/steel cooling rods with Freon flow
- Copper connectors pipe heat from chip out to cooling rods
- Freon/oil leak problem on rod construction
- Designed to keep module temperatures under 54 degrees Celsius

Floating Point

- 49 bit signed magnitude "mantissa"
- 15 bit biased exponent
- Production plans one CRAY-1 per quarter.



Topic: Vector Computers

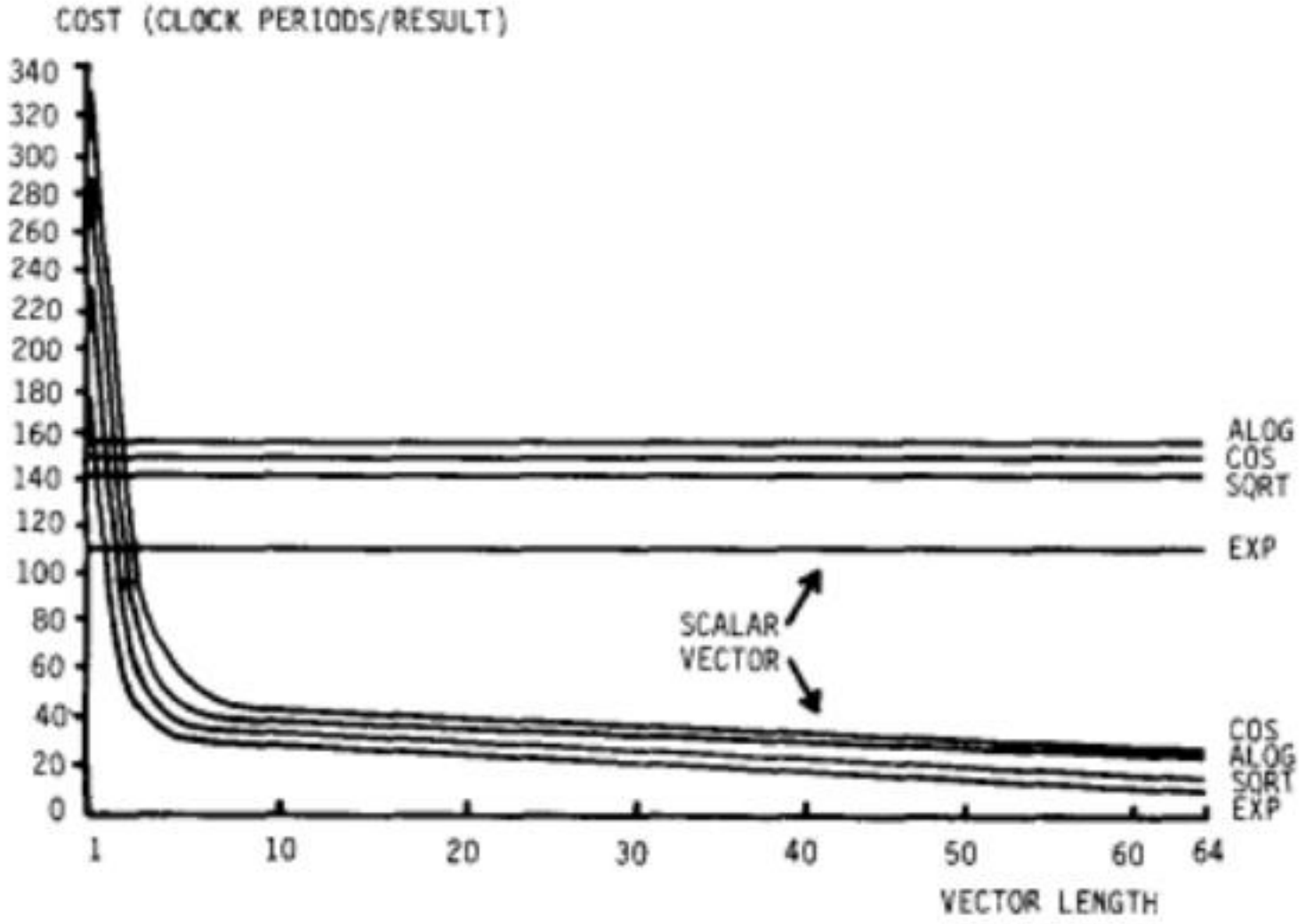
- 8 64X64 vector registers
- Process vector elements identically
- Vector Mask register can protect an element
- "Chaining"
  - Can use output of one vector operation as input to next before it is done
  - Win = don't have to store to memory then fetch from memory

Benefits of Vector Computing

- Previously needed 100+ elements for vector to be useful over scalar
  - CRAY-1 cuts that to 2-4
- Don't need to store vector elements next to each other in memory
- Max wait time is previous vector length + 4
- Common wait time is functional unit time + 2

Vector Benefits Continued

Fig. 7. Scalar/vector timing.



Compiler

- CFT (Cray Fortran)
  - Automatically vectorizes inner loop if possible
    - No need to rewrite code!
  - Can't vectorize loops with control statements.
  - Often slower than hand coded assembly.
  - Improve instruction scheduling.

Table

Table III.

Execution time in clock periods per result for various simple DO loops of the form

```
DO 10 I = 1.N
10 A(I) = B(I)
```

Loop Body	$N = 1$	10	100	1000	1000 Scalar
1. $A(I) = 1.$	41.0	5.5	2.6	2.5	22.5
2. $A(I) = B(I)$	44.0	5.8	2.7	2.5	31.0
3. $A(I) = B(I) + 10.$	55.0	6.9	2.9	2.6	37.0
4. $A(I) = B(I) + C(I)$	59.0	8.2	3.9	3.7	41.0
5. $A(I) = B(I)*10.$	56.0	7.0	2.9	2.6	38.0
6. $A(I) = B(I)*C(I)$	60.0	8.3	4.0	3.7	42.0
7. $A(I) = B(I)/10.$	94.0	10.8	4.1	3.7	52.0
8. $A(I) = B(I)/C(I)$	89.0	13.3	7.6	7.2	60.0
9. $A(I) = SIN(B(I))$	462.0	61.0	33.3	31.4	198.1
10. $A(I) = ASIN(B(I))$	430.0	209.5	189.5	188.3	169.1
11. $A(I) = ABS(B(I))$	61.0	7.5	2.9	2.6	
12. $A(I) = AMAX1(B(I), C(I))$	80.0	11.2	5.2	4.8	
13. $\begin{cases} C(I) = A(I) \\ A(I) = B(I) \\ B(I) = CCI \end{cases}$	90.0	12.7	6.3	5.8	47.0
14. $A(I) = B(I)*C(I) + D(I)*E(I)$	110.0	16.0	7.7	7.1	57.0
15. $A(I) = B(I)*C(I) + (D(I)*E(I))$	113.0	14.7	6.6	6.0	63.0
16. $A(I) = B(I)*C(I) + D(I)$	95.0	12.7	5.5	5.0	52.0

Registers

- A = 8 address registers
- B = 64 address-save registers
- S = 8 scalar registers
- T = 64 scalar-save registers
- V = 8 64X64 vector registers

Special Registers

- VM = mask off vector elements to not operate on
- VL = length of vector being processed
- P = parcel address count
- BA = absolute address used as base for indexed memory accesses (helps with dynamic user space migration)
- LA = limits the accessible address space
- XA = supports exchange operation
- F = flag register that holds various "condition codes"
- M = mode register (3 bits)
  - Bit 1 = Floating Point Error/Interrupt Enable
  - Bit 2 = Uncorrectable memory corruption Interrupt Enable
  - Bit 3 = All interrupts disabled.

Front End

- Needs an access terminal minicomputer
- Connects to a “CRAY access channel” to control the computer